MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6094

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Miniature, beam power, amplifier

Outline --- See figure 1

Base --- E9-1

Envelope --- T6-1/2 (hard glass)

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	g1	g2	h	a	h	g1	g2	k	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ehk	Ik	Pp	Pg2	TE	Alt
Unit:	V	V dc	V dc	V dc	v	mA dc	W	W	°C	ft
Maximum:	6.6	275	5, -200	275	450	75	12.5	2.0	300	(see note 1)
Minimum:	6.0	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	250	-12.5	250	0	---	---	---	---	---

GENERAL:

Qualification - Required

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Method	Requirement or test	Notes	Conditions	AQL (Percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 1								
1301	Heater current			0.4	II	If	560	640	mA
1336	Heater-cathode leakage		Ehk = +450 V dc	0.4	II	Ihk	---	25	µA dc
			Ehk = -450 V dc			Ihk	---	25	µA dc
1266	Total grid current (1)	2		0.4	II	Ic1	0	-0.5	µA dc
1256	Electrode current (1) (anode)			0.4	II	Ib	33	57	mA dc
1256	Electrode current (screen-grid)			0.4	II	Ic2	0	7.5	mA dc
1341	Power output (1)		Esig = 8.8 V ac; RL = 5,000 ohms	0.4	II	Po	3.6	---	W
1201	Short and discontinuity detection			0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes			2.5	I	R	100	---	Meg
1256	Electrode current (2) (anode)		Ec1 = -60 V dc	2.5	I	Ib	---	200	µA dc
1306	Transconductance			2.5	I	Sm	3,000	5,200	µmhos
1266	Total grid current (2)	2,3	Ef = 7.0 V	2.5	I	Ic1	0	- 2.0	µA dc
1341	Power output (2)		Ef = 5.7 V; Esig = 8.8 V ac; RL = 5,000 ohms	2.5	I	ΔPo Ef	---	15	%

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Method	Requirement or test	Notes	Conditions	AQL (Percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 2 -Continued								
1246	Audio frequency noise		Esig = 280 mV ac; Rp = 2,000 ohms	2.5	I	EB	---	17	vu
1266	Primary grid emission (screen)	4	Eg2 = 100 V ac, 60 Hz; Eb = 0	2.5	I	Ic2	---	750	µA dc
1331	Direct-interelectrode capacitance		No shield	6.5	Code E	Cg1p	1.3	1.6	pF
						Cin	7.0	10	pF
						Cout	4.0	6.0	pF
1031	Low-frequency vibration		Ec1 = -25 V dc; Rp = 2,000 ohms	6.5	S3	Ep	---	150	mV ac
1041	Shock		G = 450; Ec1 = 0; Ehk = +100 V dc; Rk = 180 ohms; Rg1 = 0.1 meg	2.5	See note 5	---	---	---	---
1031	Vibration-fatigue			2.5	See note 5	---	---	---	---
---	Shock and vibration-fatigue-test end points:								
1031	Low-frequency vibration		Ec1 =-25 V dc; Rp = 2,000 ohms	---	---	Ep	---	300	mV ac
1336	Heater-cathode leakage		Ehk = +450 V dc	---	---	Ihk	---	30	µA dc
			Ehk = -450 V dc	---	---	Ihk	---	30	µA dc
1341	Power output (1)		Esig = 8.8 V ac; RL = 5,000 ohms	---	---	Po	2.5	---	W
1266	Total grid current (1)			---	---	Ic1	0	-1.0	µA dc
1121	Base strain			---	---	---	---	---	---
2126	Envelope strain			2.5	I	---	---	---	---
1105	Permanence of marking			---	---	---	---	---	---

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Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3						
1506	Heater-cycling life		Ef = 7.5 V; 1 min "on", 4 min "off"; Ehk = 300 V ac; Ec1 = Ec2 = Eb = 0	---	---	---	---
---	Heater-cycling life-test end points						
1336	Heater-cathode leakage		Ehk = +450 V dc	Ihk	---	30	µA dc
			Ehk = -450 V dc	Ihk	---	30	µA dc
1301	Heater current			If	550	670	mA
1516	Stability life (1 hour)	6	Ec1 = 0; Rg1 = 0.5 meg; Rk = 215 ohms; TA = room	---	---	---	---
---	Stability life-test end point						
1341	Change in power output (1) of individual tubes		Esig = 8.8 V ac; RL = 5,000 ohms	ΔPo t	---	10	%
1501	Intermittent life	7,8	Ec1 = 0; Rg1 = 0.5 meg; Rk = 215 ohms; TE = +300°C (min)	---	---	---	---
---	Intermittent life-test end points (500 hours):	9					
1301	Heater current			If	550	650	mA
1266	Total grid current (1)			Ic1	0	-1.0	µA dc
1341	Power output (1)		Esig = 8.8 V ac; RL = 5,000 ohms	Po	3.5	---	W
1341	Power output (2)		Ef = 5.7 V; Esig = 8.8 V ac; RL = 5,000 ohms	ΔPo Ef	---	15	%
1336	Heater-cathode leakage		Ehk = +450 V dc	Ihk	---	20	µA dc
			Ehk = -450 V dc	Ihk	---	20	µA dc
1211	Insulation of electrodes			R	50	---	Meg
1341	Power output (1), average change		Esig = 8.8 V ac; RL = 5,000 ohms	Avg ΔPo t	---	15	%
---	Intermittent life-test end points (1,000 hours):	10					
1301	Heater current			If	550	650	mA
1266	Total grid current (1)			Ic1	0	-1.0	µA dc
1341	Power output (1)		Esig = 8.8 V ac; RL = 5,000 ohms	Po	3.2	---	W
1336	Heater-cathode leakage		Ehk = +450 V dc	Ihk	---	20	µA dc
			Ehk = -450 V dc	Ihk	---	20	µA dc
1341	Power output (1), average change		Esig = 8.8 V ac; RL = 5,000 ohms	Avg ΔPo t	---	20	%

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NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. This test shall be performed at the conclusion of the holding period.
3. Prior to this test, tubes shall be preheated 5 minutes at conditions specified below:

Ef	Ec1	Rk	Rg	Eb	Ec2
V	V dc	Ohms	Meg	V dc	V dc
7.0	0	150	0.5	250	250

4. Operate the TUT for 5 minutes at test conditions prior to the test. Adjust Ec1 to provide Ic? = 16.1 mA dc average on the positive half cycles. The grid emission shall be measured on the negative half cycles and shall not exceed the limit specified.
5. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.
6. The AQL (percent defective) shall be 1.0,inspection code H.
7. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a TUT having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.
8. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples and shall not exceed a total of three tube failures.
9. The allowable defectives per characteristic shall be not greater than 1 for the first sample and shall be not greater than 3 for combined samples.
10. The allowable defectives per characteristic shall be not greater than 2 for the first sample and shall be not greater than 5 for combined samples.
11. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
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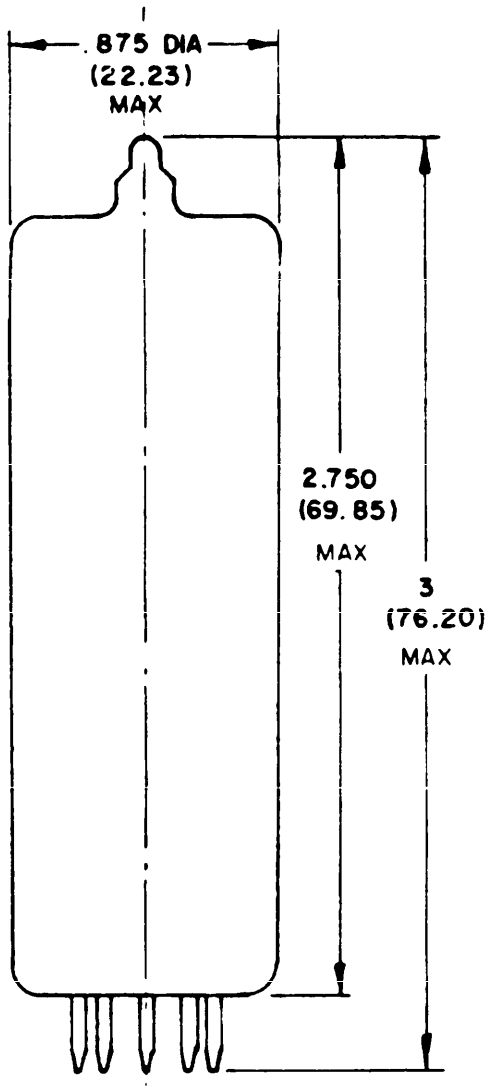
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NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
4. All dimensions to be inspected in accordance with quality conformance inspection, part 1.

FIGURE 1. Outline drawing of electron tube type 6094.

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